

3Doodler 3D Pen Class Kit

WHAT'S IN THE KIT

- 30 3Doodler 3D pens
- Charging docks
- Rainbow assortment of filament



QUICK-START GUIDE

1. Charge/warm up pens before class (~2-3 min heat time).
2. Demo: slow steady squeeze, anchor lines to the paper first.
3. Draw flat shapes on templates, then build up into 3D.
4. Filament hardens in seconds — mistakes peel off and you retry.
5. Pens are hot at the tip: one-way-to-hold rule.

FULL LESSONS IN THIS GUIDE

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PRO TIP: Template sheets under wax paper give beginners instant success — start there.

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LESSON · GRADES 3-5

2D-to-3D Shape Lift

LEARNING OUTCOMES

- Students will operate a 3D pen with safe technique
- Students will transform 2D shapes into 3D forms
- Students will use geometric vocabulary: face, edge, vertex

MATERIALS

- 3Doodler pens (charged) — 1 per student
- Shape template sheets under wax paper
- Filament assortment
- Cooling/parking spots for hot pens

INSTRUCTIONS

1. Safety brief: one way to hold, tip is hot, pens park in their spot.
2. Trace flat shapes on templates: square, triangle, circle — anchor lines to the paper.
3. Lift-off lesson: draw a second square, join the corners upward — a cube is born.
4. Shape ladder: square→cube, triangle→pyramid, circle→cylinder.
5. Count and label: faces, edges, vertices on their own builds.
6. Display the 2D-next-to-3D pairs as a geometry exhibit.

ACTIVITY

2D-to-3D Shape Lift — squares literally become cubes in students' hands. Counting faces and edges on a shape you built beats counting them on a worksheet drawing.

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LESSON · GRADES 6-8

Design a Working Object

LEARNING OUTCOMES

- Students will design a functional object to meet real requirements
- Students will identify and reinforce structural weak points
- Students will evaluate a product against its intended use

MATERIALS

- 3Doodler pens + filament
- Design planning sheet (requirements, sketch, dimensions)
- Real test objects: phones for stands, cables for holders
- Wax paper work surfaces

INSTRUCTIONS

1. Brief: build something that must actually work — phone stand, cable clip, name plate.
2. Requirements first: what weight, size, and forces must it handle?
3. Sketch with dimensions before any plastic flows.
4. Build, then test with the real object — sagging and snapping are data.
5. Reinforce weak points: triangulation and doubled strands.
6. Product demo: each object performs its job for the class.

ACTIVITY

Design a Working Object — fast, tactile product design where the phone stand must hold a real phone. Failure under load introduces reinforcement strategies naturally.

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LESSON · GRADES 9-12

Wireframe Prototyping

LEARNING OUTCOMES

- Students will translate a product concept into a 3D wireframe
- Students will analyze structural choices in a physical prototype
- Students will position wireframing within a professional design pipeline

MATERIALS

- 3Doodler pens + filament
- Sketching paper and fine markers
- Reference images of wireframe/CAD models
- Optional: Tinkercad access to model the same object digitally

INSTRUCTIONS

1. Show how designers move from sketch → wireframe → CAD → prototype.
2. Students sketch an original product concept in three views.
3. Build the wireframe: edges and structural lines only, no solid surfaces.
4. Critique pairs: where does the wireframe flex? What would a solid version need?
5. Optional: model the same object in Tinkercad and compare processes.
6. Reflection: what did the physical wireframe reveal that the sketch hid?

ACTIVITY

Wireframe Prototyping — hand-drawn 3D wireframes as the bridge between sketching and CAD. Physically feeling where the structure flexes teaches what CAD's stress views mean.